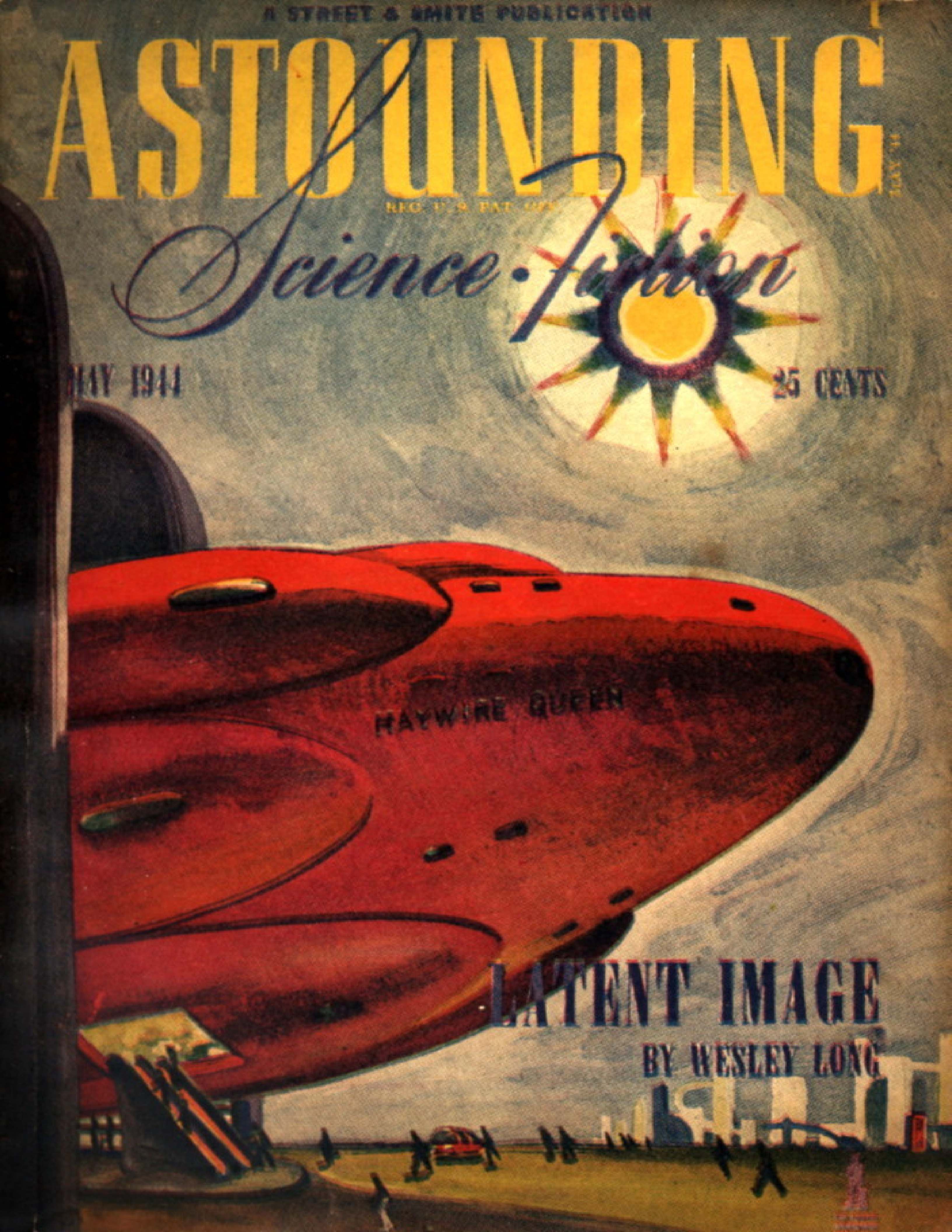
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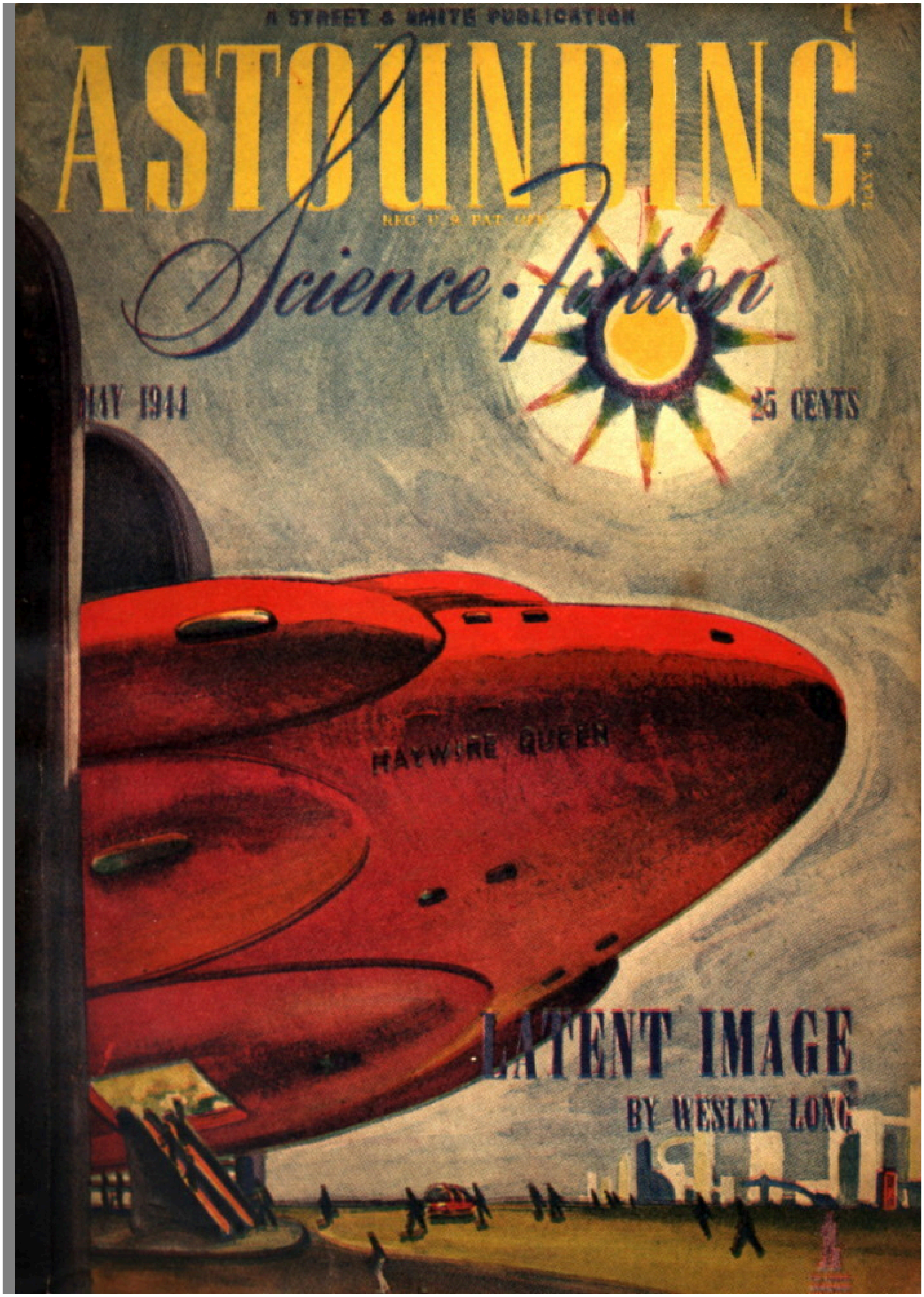
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"Latent Image"

by WESLEY LONG

Illustrated by Orban

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John McBride stood on the roof garden of Satan's Hotel, looking across the River Styx at Sharon. To his left, the River Styx emptied into the Sulphur Sea, and in the evening sky to his right, the dancing flames lighted the cloud banks over Mephisto, where the uranium smelters worked on a nonstop plan.

John McBride was in Hell.

But Hell is a city on Pluto, where the planners had a free hand because no intelligent life had ever scarred the planet until man came with his machinery and his luxury and his seeking for metal. Uranium had been found in plenty on Pluto, and so man had created a livable planet from the coldest, most forbidding planet in the System.

John McBride was in Hell, on Pluto, but his mind was dwelling in a little cube that rotated about a mythical spot halfway between Sol and Pluto; one of the many stations that created the space warp that focused Sol on Pluto with an angle of incidence equal to the incidence of Sol on Terra. Enid McBride was back there in that minute station, and John McBride wanted to be with her.

But Dr. Caldwell, the resident doctor of the Plutonian Lens, said: "John, if you've got to go to Pluto, that's O.K. But you can't take Enid with you. That's strictly out, with a capital 'O,' get me?"

"I suppose—"

"I've been doctoring for many years, John. It's safe for you to run off for a week or so, but don't move Enid. Your kid won't be born for a month, yet, but if you subject her to the 4- or 5-G you need to get from here to Pluto, you'll have—not only the baby, but as nasty a mess as you've ever seen! Take it from me, fella, 4-G is worse than a fall if you keep it up for hours. No dice!"

"O.K.," said John, unhappily. "She'll be all right?"

"Sure," said Caldwell. "Besides, all you can do now is to sit around, bite your fingernails, and ask foolish questions. If I had my way, you'd be away when the youngster is born, that'd save you from a lot of useless worry."

"That isn't fair."

"I know you feel that way. Enid does too. But it is still sort of futile. You want the right to worry; go ahead and worry. After all, there are enough people around the Lens that know you are worrying. She'll be all right, I tell you!"

"You'll let me know if anything turns up?"

"That's a promise, John."

So John McBride was standing on a roof garden in Hell, thinking how appropriate it was. He was in Hell, all right. Hell was a nice place to be, warm, pleasant, and happily balanced. But it was no place to be when your wife is nineteen hundred million miles away. Ah, well, another week of this and he would be racing homeward.

Home! That was funny, to consider home, a place in space where gravity was furnished by an mechanogravitic warp, and where there were no windows to open, and where you lived in a cube of steel three thousand feet on a side, mostly filled with the items required for living plus the maze of equipment required to maintain the great lens that gave Pluto its sun.

Home! It was a far cry from his boyhood home on Venus, where the greenery of the forest fought with the very walls. But home is where you like it, and McBride liked it.

He wished that he were there, for he felt that Enid needed him.

Then with that perversity of nature that people call fate, a bellhop approached him and handed him a spacegram. McBride tipped the boy and opened the envelope easily. He'd been getting 'grams by round numbers for several years, and this was no novelty. He was not aware of its importance until he opened the folded page and read:

> JOHN MC BRIDE
> SATAN'S HOTEL
> HELL, PLUTO
>
> HIT SKY FOR HOME. ENID IN NO GREAT DANGER FROM FALL, BUT HER RECOVERY WILL BE ASSISTED BY YOUR PRESENCE.
>
> CALDWELL.

McBride read the words twice, and then looked around himself, wildly. *Hit Sky* was easy to say—but at 6-G it would take just over one hundred hours to make the passage. Four days minimum!

McBride raced to the elevator, chewed his fingernails while the car rode him down the hundred and seven floors with that snail's pace caused by many stops. He shot out of the elevator door, caromed off the opposite wall into an ash tray which he upset and sent a small cascade of sand across the floor. McBride coasted to a stop before the hotel manager's desk and tossed the 'gram in front of him. The manager read and looked up in sympathy.

McBride said: "Get me a reservation on the next sunward-bound ship. Emergency stop; they'll make the stopoff with an emergency."

"Right." The manager spoke into the phone and then said: "And you'll be checking out?"

"Yes. Have one of the boys collect my stuff and ship it out to Station 1."

"O.K., McBride, we'll see that your stuff is taken care of. Ben!" he called out through the door, "hurry up on that reservation, and see that a car is ready to take Mr. McBride to Hellsport."

"T'won't be necessary," said Ben with a glum face. "The *Uranium Lady* just took off fifteen minutes ago, and there isn't another ship scheduled out of Hellsport for five days."

"Five days!" groaned McBride. "Anything flyable on this planet?"

"Nothing that would take a run to the Lens," said Ben.

"Sure?"

"Almost positive. However, I'll put a request on the radio that may smoke out an unknown."

"I'll buy the thing if they won't let me go any other way," said McBride.

"We understand," said the hotel manager.

McBride stamped up and down the hotel lobby for an hour. His luggage came down, all collected and prepared. He called Caldwell, and spoke to him for an hour, but Dr. Caldwell's protestations didn't help McBride. Enid had fallen from a chair while cleaning out a shelf, and was resting easily, no complications. Yes, there was some pain, enough to make Enid want her husband near. No danger, no, but it would be best if he were there.

But McBride was still one hundred hours and nineteen hundred million miles away.

John McBride didn't see the messenger boy bringing the message until he almost bumped into him. "Mr. McBride, here's your answer," said the lad, and he saw McBride rip the envelope open with a quick gesture to read the following:

> MC BRIDE:
>
> EXPERIMENTAL SPACESHIP *HAYWIRE QUEEN* AT YOUR COMMAND IF YOU CAN REPAIR ALPHATRON. MEET ME AT HELLSPORT.
>
> STEVE HAMMOND (SKYWAYS)

McBride said to the messenger: "It's grabbing at straws, but get me a cab and I'll take a whirl at it."

"Think you can do it?" asked the lad.

"Don't know. I'm desperate. After all, it's a wild chance because if Steve Hammond and his gang haven't been able to repair it, how can I expect to?"

"Give it a whirl anyway, sir," said the lad.

"That I'll do," said McBride. "And now that cab!"

The *Haywire Queen* stood above McBride as he met Steve Hammond. "What's your trouble, John?" asked Hammond.

McBride explained. Then he asked: "What's yours?"

Hammond smiled wryly. "That's a long, sad tale. We've been trying to increase the efficiency of the drive, you know. We've been hunting up and down the electrogravitic spectrum for a more efficient operating point. We found what we knew already; that we were using the most efficient part of the E-grav range. We went all the way from down low, where the stuff is just beginning to make itself detectable to up high where the equipment is slightly fragile and extremely experimental in construction. Then we took a run at the mec-grav, with absolutely no success other than to ruin a whole bank of relays; the mechanogravitic warp extended farther than we anticipated when we hit the mechanogravitic resonance of the drive bar, and hell sort of flew all over in great hunks. One of the interesting items was the closing of the E-grav field controls, and the resulting power drain over-loaded the alphatron. We limped in using a jury-rigged line from the lifeship's alphatron and made a something-slightly-less than a crash landing here on Pluto.

"So now we're either stuck here until we get the new alphatron we ordered, or you can give us a few hints on household repairs."

"What's your lifeship's output?" asked McBride, following Hammond into the spacelock.

"About eleven hundred alphons."

"You'll need about fourteen hundred to take off from Pluto," said John. "How's the big one?"

"Deader than the proverbial dodo, whatever that was."

"Dodo?" laughed McBride. "That was a mythical critter that went around dead, I think. It was so dead, even when alive, that when it really died, it was really dead."

"You'd better stick to alphatrons," laughed Hammond.

"Speaking of the equipment, have you tried to get a replacement on Pluto?"

"Nothing didding. About our only chance is to haywire something together. But remember, we still have to make a landing, somewhere, and that means a safety factor is somewhat to be desired."

"Not at all. If we can take off safely, we're in!"

"Explain. As I was taught in school, anyone can fly a spaceship, but it takes a pilot to land one."

"Sure, but remember you'll be stopping off at the Lens. We've got replacements there that will enable you to make space repairs and go on from there in safety."

"Didn't think of that. Well, here's the mess!"

McBride needed no close inspection to see that the alphatron was definitely defunct. A foul smell, faint, ephemerally pungent, permeated the room. It was the smell of burned synthetic coil dope and field-winding varnish which has been described as smelling something like a frying toupee.

"Not only dead," was his cryptic remark, "but dead and sutteed!"

"Fricasseed," corrected Hammond. "Anything we can do?"

"Is the winding intact?"

"We thought of that, too. Nope. Electrical inspection indicates that the winding is melted together in several places. You couldn't unwind the coil, let alone rewind it with fresh insulation. We've got a couple of gallons of insulation handy, if you get a good idea."

"Not yet. But look, Hammond, have you tried the magnetogravitic spectrum yet?"

"No. That was our next program."

"I'd have tried that first," mused McBride. "Knowing that the drive depends upon the action of a cupralum bar under high magnetic density plus an electrogravitic warp, I should think that the close relationship between the magnetic and electronic phenomena would lead you to try the mag-grav first."

"I didn't want to start at the top," said Hammond dryly. "In spite of the fact that Dr. Ellson claimed to have discovered a region in the mag-grav spectrum that produced a faint success."

"Well, what I'm thinking is that we can rip up the E-grav generator and use the field coil for the alphatron. It'll carry electrons as well as it carries alphons, you know."

"Better," said Hammond. "But what do we use for an E-grav?"

"First we'll hunt up through the spectrum of the magnetogravitic spectrum. If that doesn't work, we can add the warp produced by your mech-grav, run from the lifeship's little alphatron. Right?"

"It's an idea. Seems to me that I've heard somewhere that the combined warps of magneto- and mechanogravitic produces some vectors in the electrogravitic spectrum."

"Mind if I brag?" asked McBride. "That was in a paper I scribbled for the Interplanetary Gravitic Engineers. Purely a matter of making a few dimes, at the time there was nothing practical about it, since we had E-grav generators before we discovered the mechano- and magnetogravitics."

"We?" grinned Hammond. "You were still three generations in the future at the time, grandpa. But it's worth a try."

"Never thought that my effort was going to be worth a hoot," smiled John McBride. "Let's give it a whirl."

"O.K. I'll call the gang." Steve Hammond stepped to the communicator and spoke. "Jimmy, Pete, Larry! Come a-running and bring your cutting pliers!"

From what was obviously three different parts of the ship, three voices answered.

Pete arrived first. "Meet John McBride of the Plutonian Lens," introduced Hammond. "This is Pete, whose whole name is Peter Thurman, and who is the guy who knows all about drive equipment."

Pete grinned. "You see us hitting sky at two hundred feet per," he said, shaking McBride's hand.

Jimmy arrived, with Larry not far behind. "These are James Wilson and Lawrence Timkins, respectively. Jimmy is the alphatron expert, and Larry knows all there is to know about electrical circuits and wiring."

"He's ribbing me about those relays," laughed Larry, while Jimmy was saying: "Y'smell that smell? That was my pride and joy."

"Tell me," asked McBride, "what does *he* do?"

"Who, Steve? Oh he's just the bird that wanted the things done that resulted in this mess. He's primarily responsible."

"Hm-m-m. That puts the fix on the whole thing," said McBride. "Well, fellow, you've heard about Enid. I've got to get home. If we can fake up something so that the *Haywire Queen* will cut loose with a couple of hundred feet per for long enough to get me to Station 1, I'll see that your ruined equipment is replaced so that you can make a safe landing. Say! How come you do not carry a spare alphatron?"

"Why doesn't man come with two hearts?" asked Jimmy. "That's because they're usually dependable. No one ever tried to run two brains off of one heart—that's why one heart stands up pretty well. I can imagine the trouble that would result if two involuntary control centers were running the same heart—it would be something like what happened when the mech-grav made the E-grav cut in—something would blow a fuse."

They laughed, and then Hammond explained about the program. "Right away quick we'll try the mech-grav along with the mag-grav. That sounds like our best bet for something that works. Also breach the lifeship and sabotage the little alphatron for the mech-grav. Might as well have it down here where it's needed." In an aside to McBride, he added: "Is this like your place? No fuses, no safety devices, no spare equipment because some screwball is always filching something off of a bit of standard equipment to make an experimental set-up?"

"Anything but the running and operating gear of the Lens stations," said McBride, "is subject to change without notice. I've even seen a spare mech-grav generator used to counterbalance Jim Lear's teeter-totter. Jim's dad is on Station 3 and there isn't any kid of that size and age on Three. Did a good job, too, since Bob Lear fixed the mech-grav density control with a switch that urged the far end of the plank so that Jim was lifted and dropped at the right speed."

"Sort of expensive counterbalance, wasn't it?"

"I suppose so, but Bob said it was better than having to crank his son up and down by hand. Besides, we have lots of power out at the Lens." McBride paused. "Say. Do you run the *Haywire Queen* with this crew? Who's pilot?"

"Hannigan. But he got hurt when the works blew up. He ran us in all right, though any of us can take a trick at landing. But he's taking a rest cure to soothe his nerves; they got a scrambling from too much electricity."

"Too bad."

"Not so bad. Just made him jittery. He'll be all right in a week. But we won't have to run home without a pilot. I've got one coming out in a couple of hours. Drake. Ever heard of a pilot named Drake?"

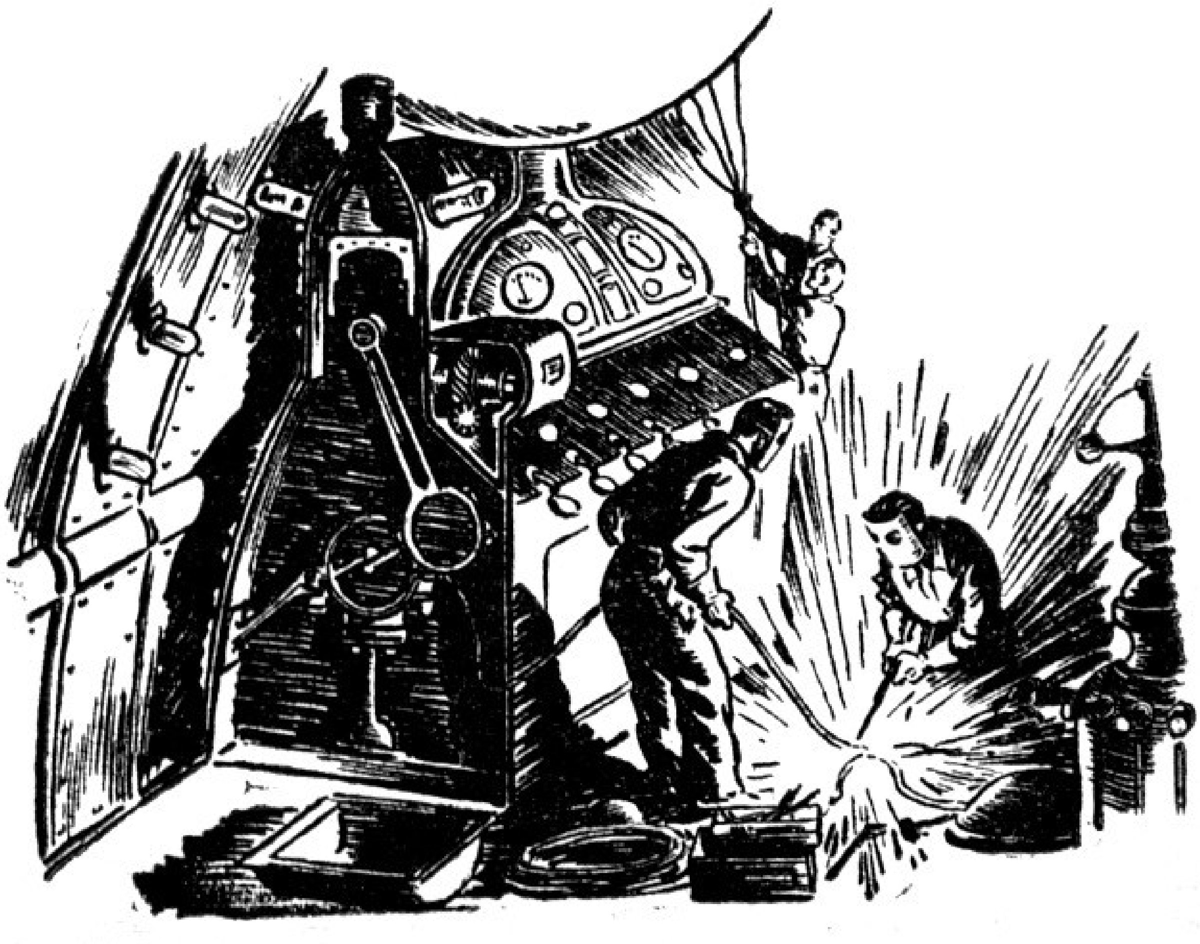
"Seems to me that the name is familiar," said McBride slowly. "But not too clear, I'll know him when I see him."

"I won't. Conducted the hiring by mail, and then gave him a call when the need came—your need, I mean. They told me that Drake was out of the building, but that he'd be at Hellsport as soon as they could find him. Has a pretty good record, too, save for one thing—"

"Steve," said one of the men, "can you give us a lift? The *Beetle's* alphatron is somewhat heavier than we can handle around this corner."

"Sure. And the next time we're at Terra, have 'em fix the hoist rail, huh?"

Wires, bunched cables, and scraps were a tangled mess on the floor. Tools were strewn about in profusion. A box of nuts and bolts had overturned and cascaded the small parts across the floor below the workbench. But the work was progressing in fine shape in spite of the seeming confusion and messiness. To someone who knew these men, it was obvious that they knew their business and how to use their tools even though the place was ankle deep in junk. To someone who knew them not, the place looked like a junk shop.



"Is this the place where the finest brains in space work out the intricate problems?" asked a cool contralto with a cynical tone.

McBride, who had just finished welding a small angle bracket on the bottom of the mech-grav generator, looked up, blinked, did a double take, and then stood up. The torch burned the air in his limp fingers, wasting the canned gas.

"You! Drake! *Sandra* Drake!"

"Is there another?" asked the saucy voice.

"I thought that Sandy was a nickname," snapped Hammond.

"It's Sandra," said she, "and it looks to me that your friend McBride is always up to his ears in junk!"

John extinguished the torch and advanced upon the picturesque red-head. "Have you still got your license?" he asked. "After that stunt you pulled—"

"Your political pals took away my private license, but I'm still registered as a pilot. This, I've been told, is an emergency, and, therefore, I am compelled to run your junk-heap for you. I'm willing for no other reason than the fact that my assistance to you in your so-called time of need will be instrumental in getting my private license back. Are you ready to go—and where?"

"We're about ready to try," said Steve.

"Try?" scorned Sandra. The perfect features twisted in a sneer. "Aren't the best brains working today?"

"Look, Pilot Drake, this is an experimental crate from way back," snapped Hammond. "You're likely to find yourself drinking coffee out of a relay-shield. We blew out the only alphatron this side of Jupiter by mishap, and John and we have been trying to gain the same effect by trusting to an experiment made several years ago but abandoned."

"I think I'll have none of it," snorted Drake. "I'd like to see a little more of the solar system before I die. You can get some other fool to run your patched-up ash can."

"Drake," said Steve Hammond, "if you do not run this crate for us—or at least try as hard as we are trying—I'll personally see that you are mentioned whenever skunks, lizards, and butyl mercaptan are talked about. This is an emergency."

"Mind telling me just what type of life-and-death run you're going for?" asked Sandra, loftily.

"Enid McBride is hurt and needs him," said Hammond, pointing at John. "There's a small matter involved—a small matter of a baby's life, possibly. If John can get there in time, his presence will give Enid the amount of lift she needs. Get me?"

"Baby?" sneered Sandra. "What woman in her right mind would have—"

"Your mother," snapped Hammond, "and she made a mistake. Now will you rectify her error and do something of value for once in your ill-used twenty-four years?"

"I've no choice," said Drake. "I'll do it. But—"

"No buts. You're under suspension right now, and how you handle the *Haywire Queen* marks your card. Take it—or take it!"

"Where's the pilot room?" asked Sandra in a cool tone.

"Below—where it usually is in a ship of this type. Your orders will be coming soon enough, I hope."

"And our destination will probably be Station 1?"

"Right. Will you need navigational details?"

"I can work them out."

Drake left, and the men put the finishing touches on the double-warp set-up. Hammond turned the equipment on, running them at test power while Jimmy and McBride adjusted the generators for maximum output.

Pete inspected the myriad of little glowing lights on the informer panel and said that the ship was working properly from dome to foot.

"Grab a rolling chair," said Hammond to McBride. Then he snapped the communicator and said: "Drake. Up at twenty feet per."

"Up at twenty feet per second per second acceleration," responded Sandra in that flat, personless voice.

"We hope," said Steve with a short laugh.

An alarm gong sounded through the open communicator, and directly afterward, the men in the power room could hear the relays closing. In the room above them, an oil switch closed with a crashing sound, its racket hardly muffled by the steel-grating floor. A rheostat whirred as it followed the impulses sent from the control board in the pilot's room; it whisked over a dozen contacts and came to rest. Four big pilot lights winked into brilliance above the informer panel, indicating that the ship was, 1.: Air-tight; 2.: Properly air-conditioned; 3.: Possessed of sufficient power for flight; and 4.: Ready to lift. Behind a two-foot dial, a diffused light glowed, illuminating the face which would indicate the acceleration in feet per second. A small dynamotor whined up the scale and into the region of inaudibility, and a series of safe lights went on; lights that would be on all the time regardless of what happened to the rest of the operating equipment. The meters of the alphatron moved slightly, and then leaped toward the top peg, stopping before they hit as the meter-sensitivity was cut accordingly. The mag-grav generator meters followed suit, and then the mech-grav meters went through the same dance. Then, far above them in the larger part of the ship, a remotely controlled tap on a bank of high-powered resistors made two steps forward, and an oil switch that connected the drive's electronic requirements to the closed-system turbine went home. Energy charged the gravitic equipment with operating power—

And the *Haywire Queen* lifted upward!

The accelerometer moved quickly up the scale toward twenty.

"We made it!" yelled Jimmy Wilson.

"We're in!" shouted Pete Thurman.

"Thank God!" breathed McBride. "I'm going to call the Lens and tell Doc Caldwell that I'm on the way—Hammond, what is that woman doing?"

The accelerometer had passed twenty, and was approaching twenty-five.

"Probably bunged the accelerometers out of sync when we crash-landed," said Hammond. "They're the standard Hooke Accelerometers, you know, and we may have stretched the spring a bit. She'll stop soon."

"It's all right," said McBride. "It just makes us get there sooner, but she shouldn't be playing with the drive this close to Pluto. If we've missed something, we'll smack."

The meter passed thirty and headed for forty feet per second per second.

"Little over one Terran G," mentioned Pete. "She probably has the usual Pilot's Fever."

"I know," agreed Hammond, "but her inherent desire to grab sky shouldn't make her play foolish with a brand-new drive."

The meter touched fifty for an instant and then went on up toward sixty. It did not stop at the green line that indicated two Terran G, but passed it and proceeded on toward seventy feet per. It climbed to eighty, passed, approached ninety, passed, and still climbed with a precise linearity that made the men admire the steady hand on the main power lever in spite of whose hand it was.

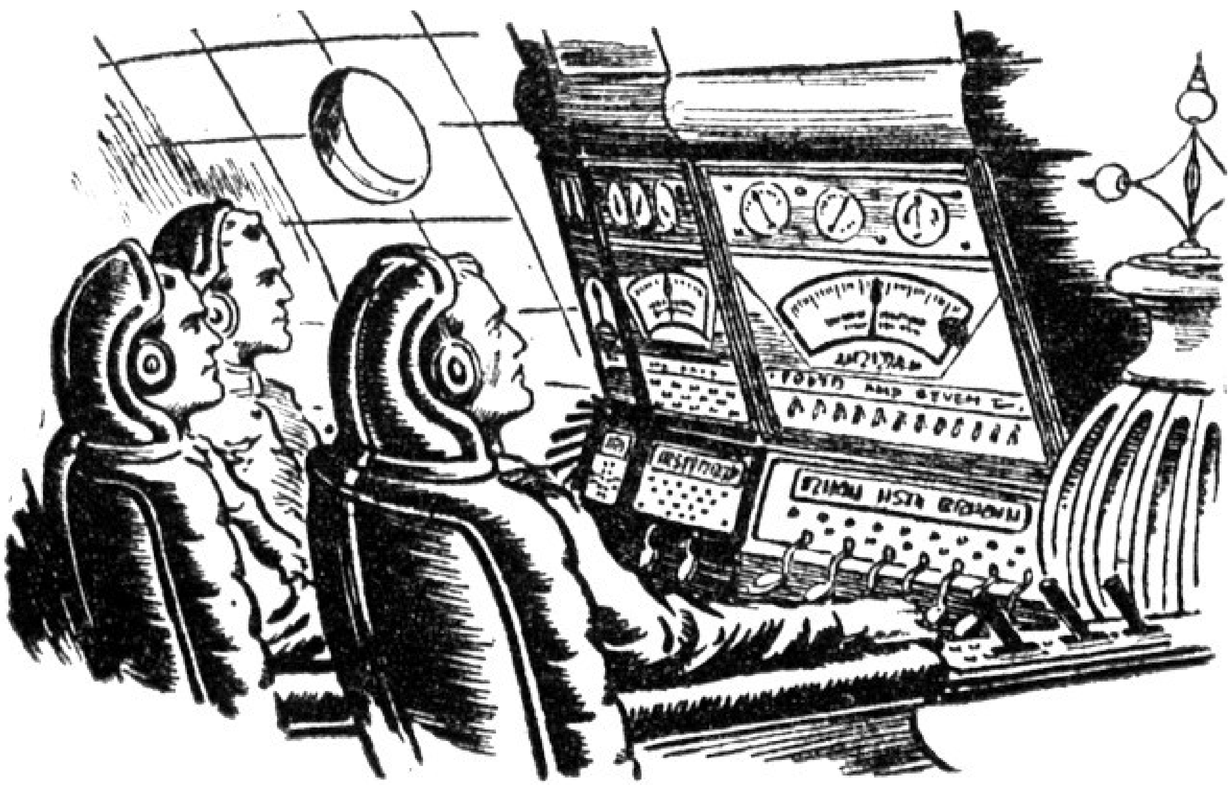
At one hundred feet per second per second, Hammond said: "When is that dame going to stop?"

"Call her down," suggested Larry.

"Better wait. No use making her nervous this close to Pluto. Bawl her out and she's likely to make the wrong move—and one move would be too much!"

The pressure of 4-G held them to their padded seats, and their heads were fixed immobile in the head braces, all watching the dial climb. It passed one-thirty, went on up the dial to one-forty, and then the voice of Sandra Drake said, weakly:

"When are you fools going to stop?"



Hammond gaped. "Who? Us?"

"Who else?" snapped Drake.

The meter touched one-fifty.

"We're not doing anything. Level off, Drake, or you'll squash us flat!"

"*You* level off. I haven't had the power lever since I adjusted it before I hit the main switch. I can't even lift my arm now; I didn't expect you to run this heap of junk from up there and so I didn't adjust the arm rest."

The dial crept up past one-sixty.

"But we're not running it from here," yelled McBride. "We haven't touched a thing since Hammond told you to take it up at twenty feet per."

The automatic respirators started to work, pumping their rib-cages in rhythm with their own breathing urges, and they sank into the enveloping folds of the elastomer cushions which supported their bodies. The meter hit one-seventy and passed it.

"Then who *is* running the soup up?" came back the labored voice of Sandra Drake. "If this is just a joke, cut it. I can't take much more."

"You—and all of us are doing well to take what we're getting now," stormed McBride. "Who—"

"Something's amiss somewhere," said Hammond thoughtfully. "At this point, any gagster would quit. Look at the meter. One hundred and ninety feet per! Almost 7-G! Uh—"

Black waves and dizziness came, shrouding little dazzles of colored pinpoints that danced before the eyes. The meter touched two hundred feet per second per second acceleration, and then the drive was cut with a snap. The compressed elastomer rebounded, almost throwing the men from their chairs, but the cover bars held them in.

The drive was completely off; acceleration zero.

"Drake! Get the inertia switch in again!" called Hammond.

"Going in," came the weakened voice of the pilot.

The original twenty feet per started again, and it began to climb, just as it did before. "Stay alive," said Hammond to Drake. "We need you to shove the inertia switch in if nothing else."

"I don't care to die," came her hard voice. "I'll keep alive. You pack of fools figure out what's wrong with your invention, that's all I ask."

"Can you crank the inertia switch down to about 5-G?" asked Pete. "Make it a hundred and fifty feet per. Then sit there and shove it in every time it comes out until we can get out of Pluto's grip. We've got to have a stable place before we can do any fixing."

"You and your jackrabbit drive," jeered Sandra Drake. "Concocted by the best brains in space. Baloney—the best space in brains, I call 'em."

"Some day," promised McBride, "I'm going to spank that woman—with a hairbrush!"

The meter rode up steadily to one-fifty, and then dropped to zero with a click. The oil switch closed again, and the meter started up the scale once again. This time McBride timed it.

"Steve," he said. "We're running at twenty per, original setting, and the acceleration is increasing at the rate of twenty feet per, also. That means our velocity is increasing at the rate of twenty feet per second per second per second."

"Something screwy. Larry, grab a few tools and ride below and fix that inertia switch so that it will close automatically. No use making Drake sit there punching on that control button every seven and one half seconds. We're going to be running this way for a couple of hours before we get to safe space."

"A couple of hours?" groaned Drake. "Listen, geniuses, is there any reason why I couldn't flatten this chair out? This is killing me."

"Look, Larry, make that switch cut out at 2-G. Sandra, set your drive for 1-G. It'll jerk our guts to pieces, but we'll be doing about the same as any ship under a 1-G drive—no, we'll be doing better. Something in this heap is making us accelerate our acceleration; we're working on the second derivative. That means—"

"That means," put in McBride, "that we're running on the rate of change of acceleration, which is the rate of change of velocity. Now under this drive, we have a new factor, which we can call 'R' and which stands for the rate of change of acceleration. Then, since our acceleration is increasing with respect to time, the linear equation: V=AT no longer holds to express our velocity at the end of T seconds. Our first equation under this rule becomes one to find the acceleration after T seconds under R rate of change of acceleration. Follow? We have Equation One:

(1) $A=RT$

Then to find the true velocity at the end of T seconds and so forth, we take the integral of that, and we have:

(2) $V=1/2RT^2$

To get the distance covered in T seconds at R rate of change, we integrate once more and come up with ... ah, let's see—Oh, sure:

(3) $D=1/6RT^3$

Is that clear?"

"I'd like to see that one worked out on a blackboard," said Jimmy. "At the present, I'll take your word for it. What I'm interested in right now is: does this factor 'R' increase with the power setting?"

"Drake just lifted it to thirty feet per," said Hammond, "and I've been timing it. So far, it does."

"Steve," said McBride, "if we can figure out some way of keeping ourselves from getting killed as the acceleration hits the upper brackets, we'll have a drive that will get us places like fury. Think fast, brother."

Hammond looked up, just as the acceleration reached a peak, and it snapped his head sharply. "*Whew*," he said. "This is fine stuff, but we couldn't run anywhere very long this way. We'd shake the whole crate loose." He was thoughtful for a minute. "Don't suppose that blackboard mind of yours could figure out our course constants from this saw-toothed curve we're running?"

"Sure," grinned McBride. "Since the thing is not increasing constantly, but is returning to zero accel each time and then building up linearly to peak, our over-all, long-time acceleration is equivalent to the average acceleration. Besides, what difference does it make? We'll get there somehow, and we can probably plot well enough to keep from doing a lot of return-chasing to hit the lens."

"O.K., but we're going to have to figure a way out of this. I couldn't stand knowing anything like this drive without trying to make it practical."

"Wait until we can talk without getting our tongues bitten off by this drive of yours, and we'll go to work on it."

McBride said: "And I'd say let's do it quick! Enid needs me—"

Sandra Drake forced the jack-rabbitting ship into a cockeyed orbit about Pluto after a couple of hours. They had nosebleed, jittery nerves, aching muscles, and voracious appetites by the time the drive was cut. They ate, smoked, took showers, and then decided to call a conference.

Hammond opened it by saying: "There's one quick way to do this. It's on everybody's tongue, but I have the floor and I'll voice it first. To keep from getting squashed by our own weight under a few thousand feet per, we need something to take up the shock—something to counterbalance the gravity-apparent. Jimmy, how many mechanogravitic generators have we aboard?"

"Two, including the one we're running in the drive right now. It is as big, however."

"How's the output of our patched-together alphatron?"

"Plenty. The coil of the E-grav was much oversize, and since electrons will rush in where alphons fear to tread, we've plenty of soup."

"Then we'll set the other mech-grav up in the nose, and extend the warp down to envelop the ship. Right?"

"Right. And now for some kind of safety factor? Supposing something goes blooey?" asked Pete. "And how do we maintain the relationship between the drive's power and the counter-drive's attraction?"

"Weight-driven power controls for the counter-drive. Inertia switches for safety. Interlocking circuits for every factor so that either the drive failure will knock out the counter-drive, or vice versa. We'll build this like an electric lock, so that the whole shebang must be right on the button before she'll move. Then the failure of any part to perform will stop all parts simultaneously. It'll probably be jerky at first, but the prime function is to get Mac to Station 1, and from there on in we can tinker with this thing until hell freezes over. O.K., let's hustle the mech-grav into the nose."

Installing the mech-grav generator in the nose of the ship was not a difficult job, since it weighed exactly nothing with the ship in an orbit about Pluto. But the intricate job of hooking the equipment together was to be more difficult.

They rammed holes in the bulk-heads to pass cables. They tore out whole sections of unimportant wiring circuits to get wire for the interlocking circuits—and when the terminals were there, the relays and inertia switches had to be made or converted from existing equipment.

Sandra Drake was of little help. She could make the ship perform to within a thousandth of an inch of its design, and perhaps add a few items that the designers hadn't included, but her knowledge of the works was small. She hadn't thought it necessary or desirable to understand, beyond the rudiments, how the drive worked.

In fact, up to the present time she had scorned the knowledge of any higher intricacies; her idea had always been that men were paid to think these things out and she was in a position to pay them for their knowledge. Let them do it, and give them hell if it was not right. Her hiring them automatically gave her the right to order them around like slaves, and since the laws that govern space travel are such that a ship's pilot or owner may demand attention to the ship, Sandra demanded such attention, needed or not.

But this was the second time in less than a year that she had seen men working with equipment. Before, it had been her fault, and she had sniffed at their labors in a scornful attitude, gaining their hatred as she had gained the dislike of so many others.

This time it was slightly different. She had been sandbagged into this job and now it seemed as though her own life depended upon the clearness of the minds of the men who worked over the equipment.

So she entered this strange world of nuts and bolts and small tools strewn around in profusion, and stood amazed at the order that was being worked out of chaos. It was apparent to her that some semblance to order must be present, since they knew where to turn to pick up the right tool, and because the right part was always less than a foot from their ready hands.

A headstrong, spoiled brat she may have been, but Sandra Drake was by no means unintelligent. After John McBride had cut off a tiny lathe-turning just in time to hand it to Pete, who seemed to need it at exactly that moment to fit into his instrument, Sandra said to McBride: "Is there any pattern to all this mess?"

"If there weren't, we'd really be in a mess." He opened the chuck, advanced the rod a few inches, and started to turn the rod down to size again.

"This," said Sandra in that infuriating voice, "is order, neatness, and efficiency?"

"We like it," answered McBride, his eyes on the cross-feed vernier. "It may not look like a drawing room, but we know what we're doing."

"Do you? Is this a sample of how the place looks every time the *Haywire Queen* goes out to experiment?"

"Undoubtedly."

"Why couldn't it be neat and clean?"

"Because we can't replace every tool back in the cabinet when we are ready to lay it down for a minute. Because it is far better to run cutting chips all over the floor and sweep 'em up once instead of running the broom every seven seconds after each chip. Because it is easier to work this way."

"Well," said Sandra, unimpressed, "the *Haywire Queen* seems deserving of her name."

"There have been a lot of ticklish space problems fought out in her," replied McBride. "Just as we're fighting one now."

"But where are your drawings? Where are your plans? Where are your calculations?"

"Our drawings will be made by draftsmen when we make the thing work," answered McBride. "No sense in having a sheaf of drawings when we'll change the thing a dozen times before it is perfected. Our plans are step-by-step, and any result from one step may change our next step. Our calculations and mathematical deductions will be handled by brilliant mathematicians who can twist simple formulas around to fit the observed data by adding or subtracting abstract terms that fit the case."

"Sounds slightly slipshod to me."

McBride cut the part from the bar and handed it to Pete. "Enough?" he asked, and Pete nodded over his shoulder.

"You can start on part two," he called.

McBride replaced the bar with a larger one and started to work it into shape. "We don't need drawings," he said. "I know what Pete wants and how they should fit together and they're fairly simple parts. He knows what he wants and knows that I know also, so why should we make a lot of sketches for something trivial?"

"It seems to me that this is far from trivial," said Sandra pettishly. "You're playing with the lives of us all."

"Your life wasn't worth a peanut when you tried to run through the lens," said McBride. "Why quibble now?"

"I lived through it," said Drake.

"You'll live through this, perhaps," said McBride. "Besides, we're not too worried about our own lives. We're all willing to take a chance on them for Enid."

"Yeah?" drawled Sandra sourly. "How about the rest of them? That's only speaking for yourself."

Steve Hammond called from across the room: "What he said still goes. He'd do as much for me!"

"Just a big bunch of mutual admirers," sneered Sandra. "Always sticking together in a pinch."

"What's wrong with that?"

"Why didn't you think of your wife a long time ago instead of worrying now. Fine show of nerves for the public consumption!"

"Miss Drake, as far as we are concerned, you haven't been properly treated. Somewhere in the Good Book is a reference to sparing the rod and spoiling the child. Do your parents twist their faces in anguish every time they see you? They should. Anyone who has foisted upon this solar system a stinking little, unfeeling rotter like yourself should hate to be alone with their thoughts. Now get out of here and let us alone."

Sandra moved back at the harshness of his voice. McBride looked behind her and instinctively put out a hand to stop her; but Sandra thought that the move meant violence and moved back faster. She collided with a dangling wire from the alphatron and went rigid. She toppled, as stiff as a board.

"Great Space!" exploded Hammond. "Jimmy, how much was that?"

"Nine hundred alphons," answered Jimmy, looking at the meters on the alphatron and making a quick calculation. "Not enough to harm. She's just had all of her voluntary nervous system paralyzed."

McBride stooped, picked her up, and carried her to a work-chair, which he kicked horizontal with his foot and dropped her into it. He went to the medicine cabinet and filled a hypo which he shot into her arm. Gradually her too-regular breathing became humanly irregular again and she moved to get up.

"Stay there," said McBride.

"Rest a bit," advised Hammond.

"And remember next time," warned Jimmy, "that this kind of a place is no place to walk backwards. Another two hundred alphons—and that is far from impossible—and you'd have been extremely dead." He wiped his forehead with a dirty cloth, mopping the beads of nervous perspiration away.

"I suppose that would have left you without a pilot," said Sandra. Her sharp remark lacked her usual conviction, however, and she realized that it fell flat. She got out of the chair and left abruptly.

"Well, I'll be—"

"Be careful," said Larry. "She isn't worth it."

"I'm going to take the bad taste out of my mouth by calling the Lens," announced McBride.

"Go ahead," offered Pete. "We'll polish off here and by the time you're through, we'll make a stab at it!"

McBride got a through connection to Station 1, and Dr. Caldwell came to the phone.

"Doc," asked McBride. "How's Enid?"

"Touch and go, lad. We're still fighting."

"Bad?"

"I'm afraid to say 'no' to that one," answered Caldwell in a tight voice.

"What does she say?"

"She's been in a coma ever since the fall, except for a minute or two in which she called for you. John, I shouldn't have sent you away."

"Don't worry about that one. After all, you didn't know she was going to take a header."

"Yeah, but—"

"You fix her up and we'll forget it."

"But suppose—"

"Doc, is it that bad?"

"I can not deny that she would be infinitely better off if you were here. She needs an emotional lift."

"I'm trying."

"I know, lad, but the next ship off of Pluto is in five days and then four more days of flight at a killing drive. Nine interminable days."

McBride debated the advisability of telling Caldwell of their experiment, but decided against it. If he said anything about the possibility of getting there sooner, Caldwell might tell Enid on the chance that it might do her some good. On the other hand, if Enid thought he were coming, and he did not come, the shock—

"O.K., Doc. We'll get there somehow."

"We'll keep fighting," said Caldwell.

He hung up the phone as Hammond spoke into the communicator. "Sunward at thirty feet per," he said.

"Thirty feet per," answered Drake. "And may we not get burned!"

"Trusting soul," observed Hammond.

Sandra thrust the main lever home with a savage motion. Deftly she juggled the steering levers until the ship pointed at Sol. "We're off," she said. "Hold your hats!"

The accelerometer climbed by the second. It hit one hundred feet per, and then slowed in its climb, approaching one twenty in an exponential curve. In the other room, a step-by-step switch continued to click off the contacts, and the generators in the turbine room whined higher and higher up the scale. Minutes passed and became a half hour.

"We're in," said McBride, with a deep exhalation. "But how in the name of sin can we tell what our acceleration is?"

"The Hooke type of accelerometer is useless when we neutralize the gravity-apparent," agreed Hammond. "We're going to have some inventing to do."

"I wonder what the limit of our acceleration is," said Jimmy. "It can't be infinite, because the mechanogravitic generator above can take only so much—"

The inertia switches went out with startling clicks, and the weight-loaded rheostats whirred home to zero. Relays danced madly as the acceleration went to zero once more.

"Right back where we started from," came the pained voice of Sandra Drake. "Can't you birds think of something practical?"

She thrust the main control home once more, hooking it up to the automatic circuit that Larry had installed. The acceleration began again. "Now we'll have some more jackrabbit drive—but with a longer jump," said Drake cynically.

"We'll have to limp all the way to the Lens on this drive," said McBride. "It isn't too good, but I can't see—"

"I'm tired of this jerky stuff," said Sandra Drake entering the room. "It seems to me that you should be able to duplicate the mess you have here by something similar up in the nose."

"Yes?" asked Steve Hammond politely. He was interested but not impressed.

"What I'm trying to say is this: Wouldn't a set-up similar to this space-eating drive also be capable of exerting mechanical attraction, thereby getting you a constantly increasing neutralizing force?"

Steve thought that one over. "Not bad. Not bad at all!"

Jimmy jumped to his feet. "It'll work, Steve. We'll have to induce the mechanogravitic force in a cupralum bar by secondary gravitic radiation, but it is a known phenomenon. Drakey, that's top!"

"Except for one thing," said Larry. "We're fresh out of magnetogravitic generators. Aside from that, we can run this heap all the way to Sirius."

Pete said: "Yeah, and if we did have one, we'd still be short a few thousand alphons. The alphatron won't carry another generator, nor will the little one upstairs." He grinned at Sandra. "We're not tossing cold water on your suggestion. It'll work—but not right now."

"Then it was good?" asked Drake with the first question of honest awe she had used in years.

"Perfect," said McBride, cheerfully. "But not quite complete. We won't censure you for that, however, since we know that you haven't been hanging around space-warp engineers for the last ten years. You couldn't have known that this mag-grav generator will do service on both ends. All we have to do is to direct the output on a two-lobe pattern instead of a single-lobe pattern, and set our induction bar up above in the field of the mechanogravitic we've already got there. Jimmy, change the output pattern of the mag-grav and we'll hike aloft with the cupralum bar." He bowed at Sandra. "Thanks to that one, we'll be moving right along!"

Pilot Drake sent the power lever home at thirty feet per, and watched the accelerator climb to exactly thirty, where it stopped and hung. Minutes passed, and the meter read constant.

Steve Hammond smiled wider and wider as the minutes added into a quarter hour. "I think our cupralum hull is helping," he said.

"How?" asked Pete.

"Why, it is collecting enough leakage-warp to create a nice large warp of its own—in which we now travel, and in which the accelerometer reads only that factor 'R' of Mac's. That meter reads the rate of change of acceleration. Drake, step it up to sixty."

Sandra advanced the drive, and the meter went up to sixty even.

"We're on the ball," said Hammond.

"We sure are," said McBride, passing a forefinger over his cheek. "It's hot in here."

"I know. And you can call the Lens and tell the Doc we're on our way."

"I tried that. The lines were busy, so I shot 'em a 'gram. They know now that we're coming."

"I wonder if your math is correct," said Steve.

"Why?"

"If it is," explained Steve, "we'll be halfway to the Lens in three hours from start—no, wait a minute. We're running at sixty feet now. That means a little better than two hours! But if they are correct, we'll be hitting almost two times the speed of light. That is not possible."

"I think we'll do it," said McBride. "After all, we're in a space warp, and no one really knows whether the laws of the universe hold in a space warp. Drake hit the Lens at about ten thousand miles per second, was stopped in time to get to one of the fore lens stations, which must have been terrific deceleration—unthinkably high—and it didn't even muss her hair. We'll know in a bit when we are supposed to hit the speed of light."

"Then for the love of Mike, what is our limiting velocity?"

"The same as any of the gravitic spectra. Gravitic phenomena propagates at the speed of light raised to the power of 2.71828—That's our limiting velocity."

"Want to make any bets?"

"I don't mind. My guess is as good as yours."

"Better," admitted Steve.

Below, in the pilot room, Sandra Drake was having a state of nerves. She was alone in the driver's seat of a ship destined to exceed the speed of light, and she was scared. For some reason, the men who professed to shy at danger were arguing the possibilities of running above the speed of light while she, who had lived the life of an adventuresome girl, a daredevil, was worrying. She listened through the communicator at their argument and cursed under her breath.

They were going at it in a pedantic manner, hurling equations and theorems and postulates at one another like lawyers with a case for the supreme court, not men who were heading for God-Knows-What at a headlong pace under an ever-increasing acceleration.

There were all sorts of arguments as to the aspect of the sky as the speed of light was approached. And how it would look at a velocity of more than light. This went on for some time, with Steve Hammond holding out for blackout and John McBride holding for a sky that crawled forward due to the angle-vectors created by the ship's passage across the light rays, until the entire sky appeared before them—all the stars in the sky would be in the hemisphere in front of them, no constellation recognizable.

"But your supposition does not recognize the doppler effect," objected Hammond. "Visible light will be out of the visible spectrum."

"True enough. But solar radiation extends from down low in the electromagnetic scale to very very high in the extra-hard UV. Visible spectra will be dopplered into the UV, all right, but the radio waves will have an apparent frequency of light, and we shall see the stars by that, I think."

"With no change in color?" asked Hammond skeptically.

"There will be a change in color, naturally. We'll observe them in accordance with whatever long waves they emit; they will in no way resemble the familiar stars we know."

"How's a poor devil going to navigate at any rate?" asked Larry. "With everything out of place—or invisible—what's he going to use for signposts?"

"In normal usage, the super drive will be fine. We've been using autopilots for years and years, setting up the whole course from take-off to the last half hour of landing. We can still do it. We'll be flying blind, but so what? We fly pretty blind as it is; no one gives a rap about the sky outside. Instrument flying is our best bet."

"Well," said Hammond, "we'll see it soon enough. The color of the stars behind us are changing right now."

"They should. We're running at three quarters of light—and, Stevey Boy, they're still visible!"

Silently they watched the sky. Dead below them, a tiny black circle appeared and the stars that once occupied this circle were flowing away from it radially. It expanded, and the region of flow spread circularly, and the bowl of the sky moved like a fluid thing towards the top of the ship until the stars at their nose were crowding together. Stars appeared there, new stars caused by the crossing of electromagnetic waves from the rear, and the sky took on an alien sight.

For a long time the stars seemed to tighten in their positions above the ship, and then the warning bell rang and the ship swapped ends easily and the bowl of the sky was below them.

Then it began to return to the fore observation point of the *Haywire Queen* as the velocity of the ship dropped. The crawl started, and the black circle diminished until it was gone. The stars continued to regain their familiar color as the *Haywire Queen* approached the normal velocities used by mankind.

Five hours after their start, the *Haywire Queen* slid clumsily to a stop beside Station 1 and made a landing. She arced a bit, since the charge-generating equipment did not have the refinements of the Lens flitters for making the ship assume the charge of the destined station. But the arc was not too bad, and within a minute after the *Haywire Queen* touched the landing deck, John McBride was knocking on the door of Dr. Caldwell's office in the hospital.

Caldwell came out of the inner door to answer the summons, and he looked up at McBride and went dead-white.

"Mac! It's you?"

"Naturally," smiled McBride. "How's Enid?"

"How did you get here?" demanded Doc.

"That's a long yarn, Doc, and it includes a whole engineering program, exceeding the velocity of light, and using a space warp as a traveling companion. How's Enid?"

"She was none too good, but we'll have her through now. Come on in!"

"First tell Tommy that the *Haywire Queen* is on the landing deck and that they're to have anything they need if we have to kill the lens to give it to them!"

"I heard that, John," said Tommy, coming in the door. "It's done." He turned on his heel and left immediately.

John approached the bedside. "Enid," he said softly.

Enid's eyes fluttered. A wave of pain passed across her face and she tried to move. McBride looked at the doctor.

"Go ahead, John," said Doc.

"Enid. I'm here. It's John."

Enid opened her mouth, gasped once, and said in a very weak voice: "John? Here?"

"Nowhere else."

"But you ... were on ... Pluto—?"

McBride thought that one over. How could he explain? He decided not to, and said: "I've been coming back for a long time, Enid. I'm here now—that's all that counts."

"Yes, John," said Enid.

"She'll be all right now," said Caldwell. "That's what she needed."

Another wave of pain crossed Enid's face, and a nurse came with a filled hypo.

Caldwell drew McBride out. "Another half hour will see her through," he told John. "You wait here and everything will be all right. I know that now, thank God."

Caldwell left McBride to re-enter Enid's room.

Steve Hammond and Sandra Drake entered the office. "How is she?" asked Hammond.

"Doc says she's going to be all right, now. I've seen her and Doc says she's perking up already."

"Good!" said Steve. "Drake, that was a nice piece of navigating. You hit Station 1 right on the nose."

Sandra felt a whole library of emotions, mixed together. She smiled a sickly smile and said: "I should have. I've been here before, remember?"

Hammond ignored the statement because he thought it sounded too much like bluster. "Drake," he said, "the *Haywire Queen* is about ready to hop for Terra. Do you feel up to running it in?"

"Steve," snapped Sandra Drake, "I'm not going to let any idiot male handle the *Haywire Queen*, and don't you forget it! After all, I'm the only pilot in the solar system that knows how to run her! I'll personally strangle both you and whomever you think you're going to get for that job, understand?"

Sandra turned and left.

"What in the name of the seven hells has got that dame?" asked McBride.

"There are a lot of ways to kill a cat besides choking it to death with cream," said Hammond thoughtfully, "but the latter way is just as effective and sometimes a lot easier. Our she-barracuda has just hit the one thing that she can't fight."

"Huh?"

"Sure. We gave her credit for doing a good job. Willing, honest credit. No matter how she may profess to despise our opinion, she can't yell 'Liar' at us because that would mean that she thought that the praise meant nothing. She's got to agree with us, or deny that she did anything worthy. And she's been living in a world of her own, trying to prove that she is the stuff. So—get me?"

"Uh-huh, I suppose so. How're you set?"

"Pretty good. We've swiped all of your spare alphatrons and a couple more gravitic generators, and we'll butter the job up a little so that we won't worry about over-loading the alphatrons. That'll take us an hour or so. How're you doing?"

"I dunno. Doc said wait here—and dammit, I'm running out of fingernails, cigarettes, and patience."

"Well hang tight. I'll be back from time to time to see how you're getting along—Hi, Doc? What's the good word?"

"It's good," sighed Dr. Caldwell.

"Honest?" yelled McBride. "Enid's O.K.?"

"Fine. From here on in it's a breeze. Oh, I forgot to tell you. She's had her son."

"She's what?" yelled McBride.

"Son. John McBride Junior, I presume. He's an ugly, carrot-colored, monkey-faced, repulsively wrinkled little monster, but Enid says he's the image of the old man."

McBride looked at Caldwell, and then rushed out to Enid's room.

"Image of the old man, hey?" asked Hammond.

"He'll develop," said Dr. Caldwell. "Junior is a Latent Image!"

THE END.

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